Though it’s not so obvious in these examples, higher entropy is also linked to higher temperature. This makes sense if you think of the difference between a block of ice and a cloud of steam. In order to be ice, the water molecules have to be arranged in a crystal structure, whereas the particles in steam are free to move around in three dimensions. But even just cooling the steam a bit reduces its entropy because the particles are moving less: they’re more constrained, or less disordered.

The important thing about entropy, in cosmic terms, is that over time it goes up. The Second Law of Thermodynamics[fn9] states that in any isolated system, the total entropy can only increase, not decrease. In other words, order does not spontaneously appear out of nowhere, and if you leave something alone long enough, it will inevitably decay into disorder. Anyone who has tried to keep their desk tidy will understand this, the universe’s most intuitive and maddening natural law.

Whether or not the universe itself counts as an isolated system can be a matter of some discussion, but taking it to be one leads us to the conclusion that the future of the cosmos is one of inevitably increasing disarray and decay. In fact, the Second Law is considered to be so inescapable and fundamental, it’s been blamed for the passage of time itself.

The laws of physics generally have no regard for the direction of time; in most situations, reversing the equations in time makes no difference to the physics. The only part of physics that seems to care at all about which direction time is going is entropy. In fact, it’s possible that the only reason we can remember the past and not the future is that “things can only get worse” is a truth so universal that it shapes reality as we know it.

“But wait!” you might say, “I completed the jigsaw puzzle! I created order! Did I just reverse the arrow of time?!”

Not exactly. The puzzle is not an isolated system, and neither are you. Technically, any local increase in entropy can be reversed with enough effort. It would be massively difficult, but you *could* unscramble an egg if you put in enough time and some incredibly sophisticated laboratory equipment. But the total entropy will always go up. In the case of the puzzle, the effort it takes you to put the pieces together requires an expenditure of energy, which means that you’re breaking down food chemicals and releasing heat and waste products (like, you know, carbon dioxide) into your environment. That heats up the room, creates particulate waste, and probably wrinkles your shirt while you’re at it. I don’t know